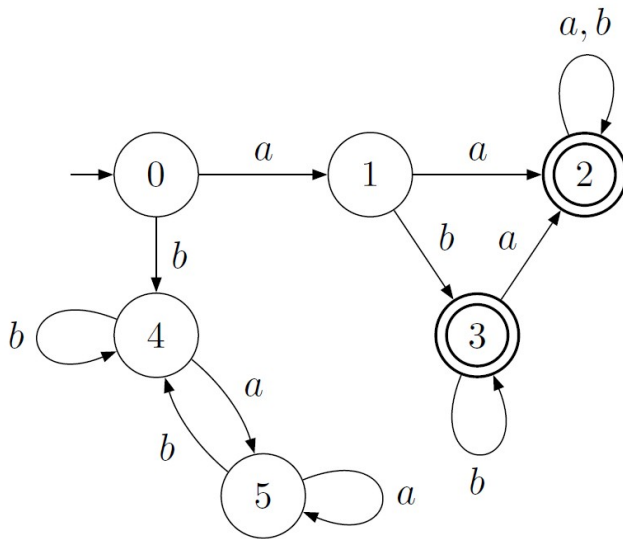




G52LAC (COMP2040)

Languages and Computation

Lecture 8: Context-Free Grammars (CFGs)

Tomas Maul
School of Computer Science
University of Nottingham Malaysia Campus

Outline

- [5 min] Questions from previous lecture?
- Context-Free Grammars.
- The language of Context-Free Grammars.

Outline

- [5 min] Questions from previous lecture?
- Context-Free Grammars.
- The language of Context-Free Grammars.

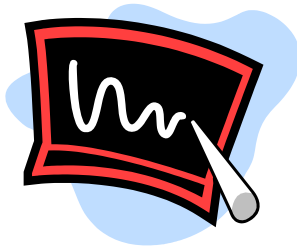
How to define Context-Free Languages?

- Pushdown Automata can express the Context-Free Languages.
- Regular Expressions are a convenient means of defining Regular Languages.
- Is there is similar way of defining Context-Free Languages?
- Context-Free Grammars!

Formal Definition of a CFG

A Context-Free Grammar is a 4-tuple (N, Σ, P, S) where:

- N is a finite set of non-terminal symbols
- Σ is a finite set of terminal symbols
- $N \cap \Sigma = \emptyset$
- P is a finite **set** of **productions**. Each production has the form $A \rightarrow \alpha$, where $(A \in N), (\alpha \in (N \cup \Sigma)^*)$
- $S \in N$ is the **start symbol**



Simple examples.
No derivations yet.

What does this mean?

Can epsilon be part of
the right-hand side of
a production?

Recommended Reading

- Introduction to Automata Theory, Languages, and Computation (3rd edition), pages 171–175 & 193–205
- G52MAL Lecture Notes, page 32

Outline

- [5 min] Questions from previous lecture?
- Context-Free Grammars.
- The language of Context-Free Grammars.

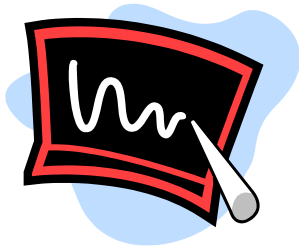
Formal Definition of a CFG

A Context-Free Grammar is a 4-tuple (N, Σ, P, S) where:

- N is a finite set of non-terminal symbols
- Σ is a finite set of terminal symbols
- $N \cap \Sigma = \emptyset$
- P is a finite **set of productions**. Each production has the form $A \rightarrow \alpha$, where $(A \in N), (\alpha \in (N \cup \Sigma)^*)$
- $S \in N$ is the **start symbol**

“Directly Derives” Relation ($\Rightarrow$)

- For any Context-Free Grammar $G = (N, \Sigma, P, S)$,
- $\Rightarrow_G \subseteq (N \cup \Sigma)^* \times (N \cup \Sigma)^*$
- If $(A \rightarrow \beta) \in P$
and $\alpha, \gamma \in (N \cup \Sigma)^*$
then $\alpha A \gamma \Rightarrow_G \alpha \beta \gamma$
- Basically: if we've got a production for A , then we can apply it regardless of context.



$A \rightarrow aA$
 $A \rightarrow zzz$
 $A \rightarrow cBff$
 $B \rightarrow bB$
 $B \rightarrow d$

Example derivations ...

“Derives” Relation ($\Rightarrow^*$)

- $\Rightarrow_G^* \subseteq (N \cup \Sigma)^* \times (N \cup \Sigma)^*$
- $\Rightarrow_G^*$ is the reflexive transitive closure of $\Rightarrow_G$
- That is, zero or more applications of $\Rightarrow_G$
- Thus $\alpha_0 \Rightarrow_G^* \alpha_n$ iff $\alpha_0 \Rightarrow_G \alpha_1 \Rightarrow_G \dots \Rightarrow_G \alpha_n$

“Derives in 1 or more steps” Relation ($\Rightarrow_G^+$)

- $\Rightarrow_G^+ \subseteq (N \cup \Sigma)^* \times (N \cup \Sigma)^*$
- $\Rightarrow_G^+$ is the transitive closure of $\Rightarrow_G$
- That is, **one or more** applications of $\Rightarrow_G$

Summary

- $\Rightarrow$ apply one production
- $\Rightarrow^*$ apply zero or more productions
- $\Rightarrow^+$ apply one or more productions

Note that when the grammar is clear from the context, the subscript is often omitted.

The Language of a CFG

- Given a CFG $G = (N, \Sigma, P, S)$, the language generated by G is defined as:

$$L(G) = \{w \mid (w \in \Sigma^*) \wedge (S \xRightarrow[G]{+} w)\}$$

- A language C is a Context-Free Language iff there exists a CFG G such that $C = L(G)$
- Reminder: The Regular Languages are a subset of the Context-Free Languages.

Sentential Forms

- Any string of terminal and non-terminal symbols that can be derived from the start symbol is a **sentential form**.
- $\alpha \in (N \cup \Sigma)^*$ is a sentential form iff $S \xRightarrow{*} \alpha$
- All words in the language are sentential forms, but not all sentential forms are words in the language.



Why?

Leftmost and Rightmost Derivations

- If a derivation always applies productions to the leftmost non-terminal of the sentential form, then it is called a **leftmost derivation**.
- The symbol $\Rightarrow_{\text{lm}}$ is used for leftmost derivations.
- If a derivation always applies productions to the rightmost non-terminal of the sentential form, then it is called a **rightmost derivation** (and the symbol $\Rightarrow_{\text{rm}}$ is used).
- $\alpha \in (N \cup \Sigma)^*$ is called a **left-sentential form** iff $S \xRightarrow{*}_{\text{lm}} \alpha$
- $\alpha \in (N \cup \Sigma)^*$ is called a **right-sentential form** iff $S \xRightarrow{*}_{\text{rm}} \alpha$



Examples

Recommended Reading

- Introduction to Automata Theory, Languages, and Computation (3rd edition), pages 175–181
- G52MAL Lecture Notes, pages 33–35

Examples/Exercises

- [A] Construct CFGs for the following languages:
 - [1] Any number of **a's**.
 - [2] One **a**, followed by any number of **b's**.
 - [3] One **a**, followed by one or two **b's**, followed by any number of **a's**.
 - [4] Any number of **a's** and **b's**.
 - [5] A sequence of n **a's** followed by a sequence of n **b's**.
- [B] CFG to derivations:
 - [1] Given the productions below, do the following words belong to the language generated by the corresponding grammar (where terminals are $\{a,b,c,d,e\}$, non-terminals are $\{S,A,B\}$ and start symbol is S): **dcbbe** and **dcbbddcbe**?

$\begin{aligned} S &\rightarrow AcBS \mid e \\ B &\rightarrow b \mid bb \\ A &\rightarrow aA \mid d \end{aligned}$
--

Acknowledgements

- Thorsten Altenkirch.
- Neil Sculthorpe.
- Henrik Nilsson.
- V.P. Kallimani.